DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Computer Science and Engineering) SEMESTER - 7 Winter 2025 (Regular)

Course :Bachelor of Technology (Computer Science and Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTCOE705B - DEEP LEARNING

Time : 3 Hours] **[Total Marks : 60**

Instructions to the Students:

1.Each question carries 12 marks.
2.Question No. 1 will be compulsory and include objective-type questions.
3.Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4.Use of non-programmable scientific calculators is allowed.
5.Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question) 12

1 Deep Learning models require
A) Less data
B)High computational power
C) Fewer computations
D) No training

2 What happens if the learning rate is set too high in gradient descent?
A) The algorithm converges faster
B) The parameters may oscillate or diverge
C) The algorithm stops immediately
D) It has no effect

3 Which optimization algorithm adjusts the learning rate for individual parameters?
A) Stochstic Gradient descent
B) Momentum Based GD
C) ADAM
D)None of the above

4 What is stochastic gradient descent (SGD)?
A) Gradient descent on the entire dataset
B) Gradient descent using one data point at a time
C) Gradient descent with a fixed learning rate
D) Gradient descent without iterations

5 What is the purpose of dropout in deep neural networks?
A) Preventing overfitting by randomly deactivating neurons during training
B) Simplifying the architecture of the network
C) Reducing the number of neurons in the network
D)Removing outliers from the training data

6 Which of the following technique resembles the autoencoders used for dimensionality reduction?
A)PCA B) Batch Normalization C)Ensemble methods D) Dropout

7 What is a filter/kernel in CNN?
A) Activation function
B) Loss function
C) Matrix used to detect patterns
D)Regularization Method

8 In denoising autoencoder noise is applied to
A)Encoder's input B)Decoder's input
C)Encoder's output D)Decoder's output

9 What is bias in machine learning?
A)Error due to overly simplistic assumptions
B)Error due to noise in the data
C)Error from excessive complexity
D)None of the above

10 Increasing the number of hidden units generally:
A)Reduces model capacity
B)Increases representational power
C)Guarantees perfect accuracy
D)increases bias

11 Which of the following neural network is used for handling time-series data?
A) RNN (Recurrent Neural Network)
B) CNN (Convolutional Neural Network
C) GAN (Generative Adversarial Network
D) Autoencoder

12 Which of the following is a major advantage of LSTM networks?
A) Ability to capture long-term dependencies
B) Reduced computational complexity
C) Enhanced feature extraction from images
D) Improved training speed

Q2. Solve the following.

A) Explain the perceptron model .What are the limitations of perceptron model? 6

B) Explain Gradient Descent algorithm for training a neural network 6

Q3. Solve the following.

A) Explain Principal Component analysis for dimensionality reduction. 6

B) Explain the backpropagation learning in detail. 6

Q4. Solve Any Two of the following.

A) List and explain the various activation functions used in modeling of artificial neuron.A neuron receives inputs x=[2,1]weights w=[0.3,0.6] and bias b=0.2. Compute the neuron output with ReLU activation. 6

B) What is regularization? Explain L1 and L2- regularization 6

C) What are autoencoders ? Explain in detail. 6

Q5. Solve Any Two of the following.

A) Explain working of all the layers in Convolutional Neural Network 6

B) Differentiate between LeNet and AlexNet architecture 6

C) Discuss some key applications of Convolutional Neural Network and Artificial Neural Network. 6

Q6. Solve Any Two of the following.

A) Explain Recurrent neural network architecture in detail. 6

B) How Deep learning differs from machine learning.Explain in detail. 6

C) Write a note on i)Data Augmentation ii) BatchNormalization 6

*** * * End * * ***